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Fault diagnosis of rotating machinery using a signal processing technique and lightweight model based on mechanical structural characteristics

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ABSTRACT

Intelligent Electromechanical Actuator (EMA) is becoming a key direction for future development, which increases the demands on the performance and complexity of fault diagnosis methods. Rotating machinery, the key rotating load-bearing components in EMA, directly impacts operational status of the entire machinery equipment. Operating in harsh environments and prone to noise, it requires fault diagnosis methods with robust noise resistance. Most existing fault diagnosis methods focus on Gaussian white noise, overlooking the impact on rotating machinery in real industrial scenarios. Additionally, they emphasize signal processing and neural networks, neglecting mechanical structure characteristics and lacking interpretability. To address these limitations, an interpretable fault diagnosis method of rotating machinery based on mechanical structural characteristics, called rolling element separation and channel-sharing, spatial-unidirectional convolutional network (RES-CSSUC), is proposed. This method integrates the rolling element separation (RES) signal processing technique and a lightweight model called CSSUCNet, which is mainly composed of channel-sharing, spatial-unidirectional convolutional layers. To assess the ability of the proposed method extracting features in noise conditions, Gaussian white noise and impact noise are respectively injected into the data. This method is demonstrated on Paderborn University bearing dataset and Planetary Roller Screw Mechanism (PRSM) dataset. Results show that RES-CSSUC achieves high diagnostic accuracy both in noise-free and noise conditions with fewer parameters, fewer FLOPs, and less memory usage, demonstrating strong robustness. It can reduce computational and storage costs, making it highly suitable for integration with mechanical equipment and facilitating the development of intelligent EMA.

1. Introduction

At present, the development of fully electric or more-electric equipment has become a trend. As a key component, the Electromechanical Actuator (EMA) shown in Fig. 1 is receiving increasing attention and has already been applied in numerous fields, such as aircraft control surfaces, automotive steering and braking, ship rudder steering, presses, riveting and gluing in the automotive manufacturing industry, coal mining equipment, robotic linear joints, and more. As the main rotating load-bearing components, the motor, bearings, and PRSM directly affect the operational status of the EMA and are prone to failures [1,2]. To reduce economic losses or even potential injuries caused by EMA shutdowns due to faults in these three components, and to enable maintenance personnel to quickly locate faults, condition monitoring is necessary. For the motor, the simplest way to monitor its condition is by measuring the current without the need for additional sensors. However, for the bearing and PRSM, it is usually necessary to use an accelerometer to capture their signals for condition monitoring. In response to these needs, the rise of intelligent equipment is driving the integration of equipment, sensors, and fault diagnosis systems. This integration facilitates real-time monitoring of EMA operational status, making it possible to detect and address issues promptly and improve overall reliability [3]. Fault diagnosis system is typically deployed on edge device, which has quite limited computing and storage resources [4]. In order to reduce storage and computational costs, an interpretable fault diagnosis method with a lightweight model based on mechanical structural

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characteristics is necessary for rotating machinery.

The vibration signals of rotating machinery contain abundant feature information [6]. It is easily integrated with rotating machinery. It can achieve early rolling bearing fault diagnosis in induction motors [7] and railway [8]. Additionally, it can also reflect the states of helicopter Tail-Drive system [9] and turbine [10] and so on.

For the samples in the above methods, signal processing is required before feeding them into subsequent neural network models, such as empirical mode decomposition, Fourier transform, and ensemble empirical mode decomposition. These methods are used to extract features from signals. In addition to these methods, there are other signal processing methods, including feature scaling [11,12], resampling [13], data augmentation [14,17] and so on. Scaling the raw signal to a common range can meet the requirements of the fault diagnosis method and accelerate the convergence speed of the model. Resampling can increase or decrease the number of samples, achieving the goal of data augmentation. It can also help to reduce the experimental time and cost associated with data collection. It changes the sampling frequency to some extent. In addition to resampling, data augmentation can increase the number of samples. It can also mitigate the data imbalance problem [14,15] and improve training stability. Data augmentation can be achieved through various methods, such as signal processing [13], incorporating the Convolutional Block Attention Mechanism module [16], and using generative adversarial networks [17].

In addition to the aforementioned literature, there are some methods that do not require feeding samples into a neural network. These methods can achieve fault diagnosis solely through signal processing techniques. For example, Li et al. [18] proposed maximum noise ratio kurtosis product deconvolution (MNRKPD) to effectively extract faults features in low signal-to-noise ratio of vibration status signals. Through using nonlinear chirp components, the effect of the structure vibration transfer paths can be removed and the quality of vibration signals can get improved. [19]. Decomposing the signal into a series of independent components containing the main fault information can reflect the states of rotating machinery [20]. In reference [21], a novel Drum-shaped Metastructure (DMS) is proposed to achieve high signal-to-noise ratio fault information by enhancing weak bearing fault signals. For these methods, it is essential to develop accurate physical models. However, physical models are usually very complex, so this often requires specialized expertise and a considerable time commitment. These methods may not scale well to large or highly complex systems with numerous interacting components. Simulating physical models can be computationally intensive, which may not be suitable for real-time fault diagnosis. Moreover, all these signal processing methods solely focus on signal processing techniques, without integrating the mechanical structure of rotating machinery. Given these limitations, there is a growing need for more advanced approaches. Deep learning models, which have demonstrated excellent performance in computer vision domains, are now being leveraged to enhance fault diagnosis. By embedding physically meaningful time–frequency transform (TFT) into the traditional convolutional layer, Chen et al. [22] improved the diagnostic performance and revealed the logical foundation of the CNN prediction in a frequency domain view. This integration opens up a wide range of possibilities for improving accuracy and efficiency. Furthermore, graph neural networks have made monumental achievements in the field of intelligent diagnosis. It can effectively handle the challenges posed by signal discrepancies across different channels [23]. In addition, graph neural networks exhibit strong adaptability. For example, it can be combined with the anhysteretic evolution learning mechanism for fault diagnosis in transmission systems [24] and it can also integrate multi-source acoustic and vibration signals, enhancing the representation of these signals [25]. Graph neural networks have also achieved good performance in the field of unsupervised fault diagnosis. In reference [26], graph neural networks effectively overcome the limitations of label dependence and improve diagnosis performance by fully mining internal correlations between samples and integrating deep representation features.

In computer vision domains, large models have been demonstrated excellent performance, such as GoogLeNet [27], Vision Transformer (ViT) [28] and so on. However, large models are not suitable for devices with limited computational power and storage resources. In some scenarios such as mobile devices, there is a necessity to utilize lightweight models, such as MobileNetV3 [29], EfficientNetV2 [30], and so on. In the field of fault diagnosis, lightweight model is also crucial for micro-industrial computers. Li. et al. [31] proposed an effective fault diagnosis method, termed as WaveletKernelNet (WKN), where a continuous wavelet convolutional (CWConv) layer was designed to replace the first convolutional layer of the standard convolutional neural network. Yu. et al. [32] developed a lightweight model named class-weighted contrastive learning quadratic network (CCQNet) in the field of fault diagnosis. Yan et al. [33] developed a lightweight model called LiConvFormer by designing a separable multiscale convolutional block and a broadcast self-attention block, which is simpler and can be applied to a wider range of industrial scenarios. Most lightweight fault diagnosis models focused on the design of neural networks, yet they did not take into account the structural characteristics of rotating machinery.

The vibration signals of rotating machinery in EMA are often influenced by noise in real industrial scenarios. Therefore, it is essential to evaluate the noise-resistant performance of fault diagnosis methods in noise conditions [34]. Gaussian white noise exists at any stage of the original vibration signal [35]. In different industrial environments and operating conditions, the intensity of Gaussian white noise in the signal varies. Therefore, in many studies, Gaussian white noise of varying intensities is injected into the signal to evaluate the performance of the proposed fault diagnosis method. For example, in reference [36], Gaussian white noise with SNR levels of -5 dB to 5 dB is added into signal. To simulate heavy Gaussian white noise conditions, SNR levels from -6 dB to 6 dB is added into samples to simulate real industrial implementation [37]. In order to evaluate the robustness of the fault diagnosis method, low-intensity Gaussian white noise can be used to

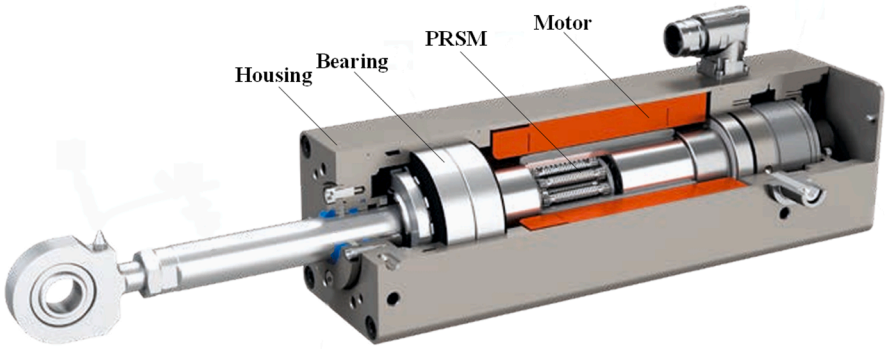


Fig. 1. The structure of EMA [5].

train the model, and high-intensity Gaussian white noise can be used to test the model. For example, Li et al. [38] trained the model for a noise level ranging from −5 dB to 5 dB and evaluated the model for a noise level ranging from −8 dB to 8 dB. In addition to Gaussian white noise, the vibration signals of rotating machinery in EMA may also be subject to interference of impact noise [39], such as cars bumping on rough roads, ships swaying in waves, airplanes experiencing turbulence due to airflow, and so on. However, there are almost no researches examining how impact noise affects the performance of fault diagnosis methods.

Based on the above analysis, a new fault diagnosis method of rotating machinery using a signal processing technique and a lightweight model based on mechanical structural characteristics is proposed, and the main contributions are summarized as follows:

1) A novel signal processing technique – rolling element separation (RES) is proposed based on the structural characteristics of rotating machinery. This technique transforms vibration signals into one-channel two-dimensional matrices, which serve as the inputs for subsequent neural network model.

2) A lightweight neural network layer, the channel-sharing, spatial-unidirectional convolutional layer (CSSUCLayer) is designed for feature extraction and fault classification of rotating machinery. CSSUCLayer reduces the number of parameters and memory consumption. By replacing the convolutional layers in ResNet18 with CSSUCLayers, a lightweight model called CSSUCNet is developed for fault diagnosis of

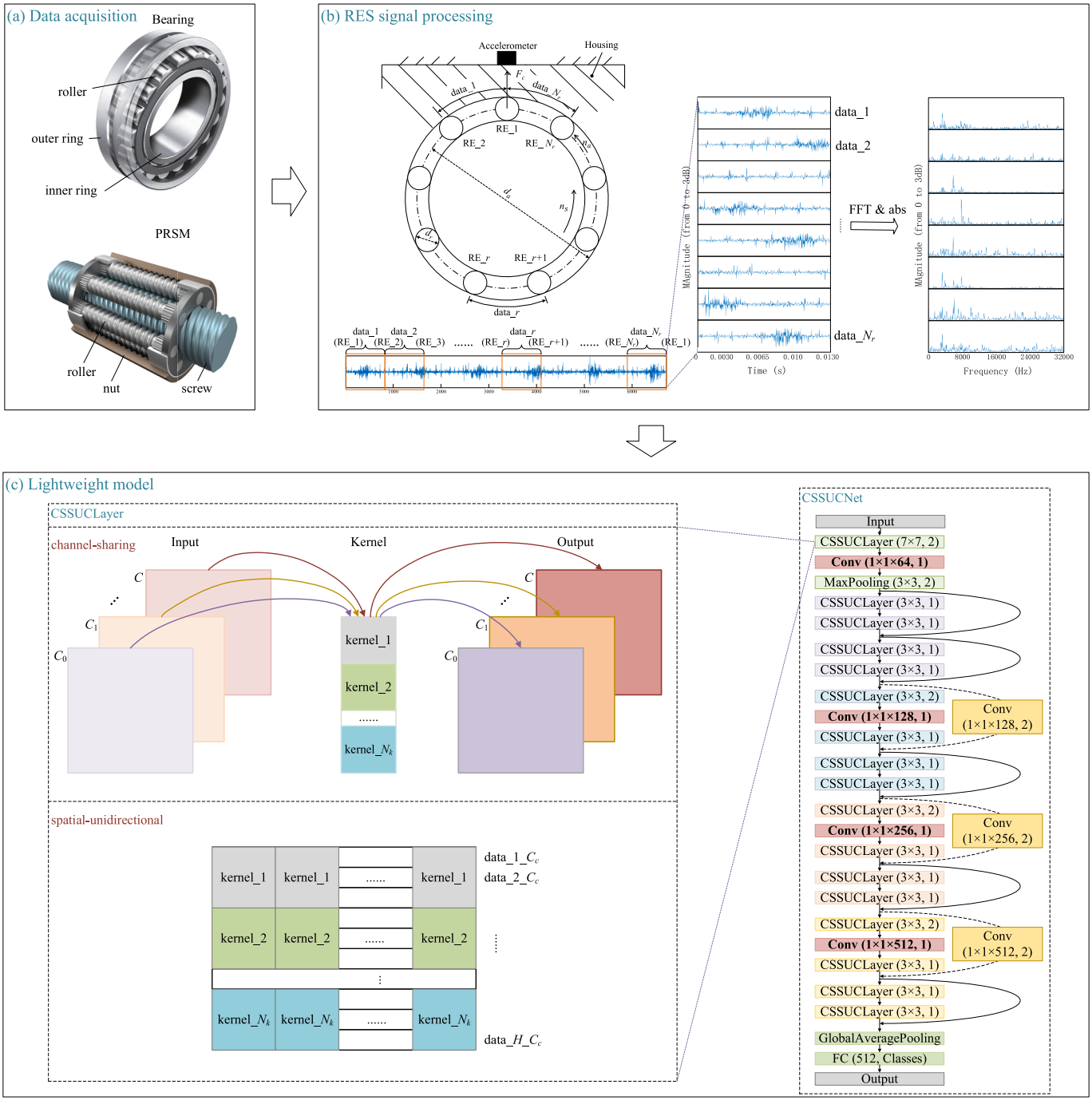


Fig. 2. The overall framework of the proposed method for fault diagnosis of rotating machinery: (a) data acquisition for rotating machinery; (b) RES signal processing technique; (c) CSSUCLayer and the architecture of CSSUCNet.

rotating machinery.

3) Two datasets of two types of rotating machinery are constructed. One is from Paderborn University bearing dataset, while the other is from our laboratory and the object is Planetary Roller Screw Mechanism (PRSM). To evaluate the anti-noise ability and the performance of the proposed method in noise conditions, Gaussian white noise and impact noise are respectively injected into simples. The experimental results indicate that the proposed method achieves high diagnostic accuracy with lower cost. Moreover, it can maintain great performance in various noise conditions.

The remainder of this paper is organized as follows. Section 2 introduces the detailed procedure of the proposed fault diagnosis method. Section 3 covers the experimental procedure. In Section 4, experiments are conducted on two cases of rotating machinery to verify the effectiveness of the proposed method, with the results and discussions provided. Conclusions are provided in Section 5.

2. Methodology

This section provides a detailed description of the proposed fault diagnosis method, referred to as RES-CSSUC, with its overall framework illustrated in Fig. 2. In this framework, two key techniques are proposed, i.e. (1) a rolling element separation (RES) signal processing technique, and (2) a channel-sharing, spatial-unidirectional convolutional layer (CSSUCLayer). Both of these techniques are based on the structural characteristics of rotating machinery. Through RES signal processing technique, one-channel two-dimensional samples are obtained as the inputs for neural network model. Then, CSSUCNet with fewer parameters, low computational complexity, and fast inference speed is used to extract features and establish nonlinear relationships between samples and fault classification. The proposed method in this paper overcomes the limitation of feature scarcity in one-dimensional data and the difficulty of one-dimensional convolutional neural network model in capturing long-term interval signal features. Additionally, it addresses the drawback of feature redundancy in the channel domains and the inability to accurately capture features in the spatial extent for two-dimensional convolutional layers.

2.1. RES signal processing

Rotating machinery generally consists of three primary components: inner ring, outer ring, and rolling elements. However, depending on the type of rotating machinery, the nomenclature for these components varies. For instance, in the case of PRSM, the inner ring, outer ring, and rolling elements are respectively referred to as screw, nut, and rollers, as illustrated in Fig. 2(a). In the operation of rotating machinery, rotating machinery typically exhibits a common mechanical behavior: the outer ring remains stationary, while the inner ring rotates. The rolling elements not only revolve around the inner ring but also spin on their own axes. Moreover, as show in Fig. 2(b), accelerometer is usually installed on the housing, and the outer ring of the rotating machinery is typically fixed to the housing (for PRSM, the accelerometer can also be directly mounted on the outer ring). This dual motion results in a complex interaction between the mechanical components, which directly influences the vibration signals that are generated during operation.

The placement of an accelerometer on the housing (nut), as shown in Fig. 2(b), plays a crucial role in capturing these vibration signals. Whenever the rolling elements pass by the position where the accelerometer is mounted, the centrifugal force F_c generated by the rotation of the rolling elements is directed towards the accelerometer. At this moment, the rolling elements are closest to the accelerometer, which causes an impact on the accelerometer. The faster the speed, the more pronounced the impact. This impact creates fluctuations in the recorded vibration signal, contributing to a unique pattern that reflects the condition of the machinery. Although centrifugal forces also exist at other positions of the rolling elements, the direction of these forces does not point towards the accelerometer, and the distance is relatively larger. Therefore, the majority of the impact generated at these positions cannot be transmitted to the accelerometer. As a result, the impact caused by the rolling elements at other positions has a minimal effect on the characteristics of the signal collected by the accelerometer. Due to the typically high operational speeds of rotating machinery, the time interval between consecutive impacts (i.e., between two adjacent rolling elements) is extremely short. This leads to the following key observation: when the initial position of any rolling element aligns with the position of the accelerometer as shown in Fig. 2(b), there will be minimal or no significant change in the signal characteristics between the passing of two adjacent rolling elements. This phenomenon arises because of the near-identical interactions between consecutive rolling elements and the stationary outer ring.

Given the structural and operational characteristics of rotating machinery, a novel signal processing technique, termed the RES, is proposed in this paper. This technique represents the vibration signal as a two-dimensional matrix, which captures the repetitive nature of the rolling elements' interaction with the outer ring and housing.

The vibration signal is modeled as an $N \times W_0$ matrix, where N represents the number of rolling elements and W_0 represents the number of data points when the rolling elements rotate $360^{\circ}/\mathrm{N}$ around the inner ring. This matrix structure reflects the periodic passage of rolling elements across the accelerometer, offering a more organized and interpretable format for subsequent analysis. The process is illustrated in Fig. 2(b). Firstly, the rotational speed of the rolling elements around the inner ring, denoted by n_a, is a key parameter that governs the time intervals between impacts. It can be calculated as follows [40]:

$$\mathrm{n_a} = \frac{\mathrm{n_s}}{2}\left(1 - \frac{\mathrm{d_r}}{\mathrm{d_a}}\cos\alpha\right) \tag{1}$$

where $\mathrm{n_s}$ is the rotational speed of the inner ring, $\mathrm{d_r}$ is the diameter of the rolling element, $\mathrm{d_a}$ is the diameter of the rolling elements revolving around the inner ring, and α is the contact angle between the rolling element and the raceway.

Secondly, when the rolling elements revolve around the inner ring for one complete revolution, the total number of data points collected by the accelerometer is denoted as L, which can be expressed by the following equation:

$$\mathrm{L} = \frac{60 \cdot \mathrm{Fs}}{\mathrm{n_a}} \tag{2}$$

where Fs represents the sampling frequency of the accelerometer.

Thirdly, to focus on the signals captured by the accelerometer during the passage of each rolling element, the signal is divided into sections corresponding to the angular displacement of $360^{\circ}/\mathrm{N}$. When the rolling element rotates $360^{\circ}/\mathrm{N}$ around the inner ring, the number of data points collected during this time interval, denoted as W_0, is obtained according to Eq. (3).

$$\mathrm{W_0} = \frac{\mathrm{L}}{\mathrm{N}} \tag{3}$$

Thus, each row of the matrix corresponds to the signal generated by the passage of one rolling element as detected by the accelerometer.

Finally, the vibration signal of length L is represented into the $N \times W_0$ matrix, the next step is to apply the Fast Fourier Transform (FFT) to each row of the matrix, followed by taking the absolute value of the result. The FFT is used to transform the time-domain signal into the frequency domain, revealing the underlying periodic components that may be indicative of faults or anomalies in the machinery. This frequency-domain representation is crucial for neural network-based fault diagnosis models, as it provides a more comprehensive view of the signal characteristics. The resulting matrix, with FFT applied, serves as the input for the neural network model, enabling it to learn the intricate patterns associated with healthy and faulty states of the

rotating machinery.

The RES signal processing technique can resist noise in the signal to some extent. Its noise resistance stems from its structured approach to capturing the periodic nature of the rolling elements' interactions with the outer ring and housing. By transforming the time-domain signal into a two-dimensional matrix, RES signal processing technique inherently filters out non-periodic components that often correspond to noise. Noise, particularly in rotating machinery, tends to manifest as random fluctuations that do not align with the periodic behavior of the machinery components. This separation between periodic signal and non-periodic noise allows RES signal processing technique to better isolate and highlight the mechanical signal of interest.

The feature extraction process in RES signal processing technique works by taking advantage of the repetitive mechanical interactions between the rolling elements and the outer ring. By constructing an $N \times W_0$ matrix, where each row corresponds to the signal generated by the passage of a rolling element, the technique emphasizes the periodicity of the mechanical impacts while suppressing random noise. When FFT is applied to each row, the transformation shifts the focus from time-domain noise to the frequency-domain characteristics. Random noise often lacks a consistent frequency pattern and is distributed across various frequencies, whereas mechanical faults in rotating machinery tend to generate specific frequency signatures. FFT helps distinguish these fault-related frequency components from background noise.

It should be noticed that only when the initial position of any rolling element coincides with the installation position of the accelerometer as shown in Fig. 2(b), the signal between two adjacent rolling elements will not contain impact vibrations, because no rolling element passes over the accelerometer within this interval. However, in most scenarios, the initial position of the rolling element does not coincide with the installation position of the accelerometer. The relationship between the position of the accelerometer and the initial position of the rolling element can be seen in Fig. 3. Assuming that the r-th rolling element is represented as RE_r and the position of RE_r is the initial position of the rolling element. The angle between RE_r and the accelerometer is denoted as β and the angle between two adjacent rolling elements is denoted as $\beta + \theta$. Since the rotating machinery can stop at any position during operation, making β and θ potentially any values. Moreover, rolling elements are typically enclosed within the rotating machinery, making it impossible to measure the values of β and θ. Based on this situation, the values of β and θ for each sample remain unknown. Therefore, it is necessary to analyze the influence of β and θ on signal segmentation.

There are four scenarios and they are listed below.

(i) $\beta = 0$, the initial position of RE_r coincides with the installation position of the accelerometer, which is the same as the situation in Fig. 2 (b).

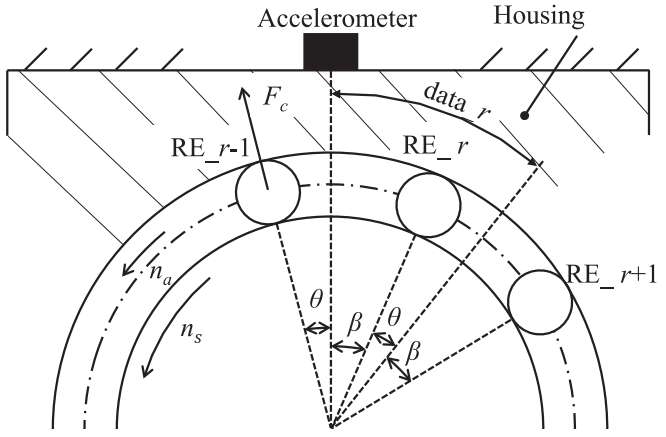


Fig. 3. The relationship between the installation position of the accelerometer and the initial position of rolling elements.

(ii) $\beta < \theta$, in this scenario, when the rolling element rotates 360°/N around the inner ring, RE_r will cause impact on the accelerometer, and the majority of data points in the signal occur before the impact.

(iii) $\beta = \theta$, when $\beta = \theta$, RE_r will also cause impact on the accelerometer. However, in this scenario, the number of data points before the impact is equal to the number of data points after the impact in the signal.

(iv) $\beta > \theta$, this scenario is similar to the scenario of (ii). The difference is that the majority of data points occur after the impact.

In scenario (i), when rolling element rotates 360°/N around inner ring, no significant impact is generated, resulting in minimal change in the signal characteristics. In scenarios (ii), (iii) and (iv), the signal is divided into two segments: before and after impact. In scenarios (ii) and (iv), majority of data points occur either before or after impact, meaning most of the data points come from the same segment of the signal, as a result, the majority of data points exhibit similar signal characteristics. Additionally, since the rotating element rotates 360°/N around inner ring for a very short time, the difference in characteristics between the two segments remains minimal. In scenario (iii), although the number of data points before and after impact is equal, the short time interval results in only minor changes in the characteristics between these two segments of the signal.

Faults in rotating machinery are generally categorized into continuous faults, such as lubrication failure, wear, and intermittent faults, such as pitting, cracks. Continuous faults typically occur throughout the entire or most of the phase which the rolling elements revolve around the inner ring for one revolution. Therefore, most of the continuous data points are fault data. This situation is similar to the scenarios (i)-(iv) described above, where the signal characteristics between two adjacent rolling elements are considered to there are no significant changes. In contrast, intermittent faults, such as pitting on a specific rolling element or when a normal rolling element passes over a pitted or cracked inner or outer ring, impact is generated. In this situation, the signal between two rolling elements will contain impact vibrations, meaning that the signal characteristics may change. However, since intermittent faults typically occur at only one location during a single rotation of the rolling element revolving around the inner ring, the signal can be divided into two main parts: before and after the fault, which is similar to scenarios (ii) and (iv).

2.2. Lightweight model

Convolutional layers have been proven to have excellent feature extraction capabilities in the field of computer vision in recent years [41–43]. However, in the field of fault diagnosis, signals are often one-dimensional time-series signals. Therefore, two-dimensional convolutional layers in the field of computer vision may not be fully suitable. With this characteristic, many fault diagnosis methods utilize one-dimensional convolutional layers to extract features [44–46]. To some extent, this can reduce a significant number of parameters. For time-series signals of rotating machinery, the signal length is often long. However, due to limitations in computational resources and model complexity, the size of one-dimensional convolutional kernels is typically limited. Additionally, for one-dimensional convolutional kernels, there is only a fixed set of parameters in the spatial extent, which means they can only capture short-term signal features, struggling to capture long-term signal features. In response to the aforementioned challenge, a novel feature extraction layer and network with fewer parameters has been proposed.

2.2.1. CSSUCLayer

As shown in Fig. 2(c), CSSUCLayer is composed of channel-sharing and spatial-unidirectional convolution. The key motivation behind channel-sharing convolution is to address the redundancy that often exists across different channels of convolutional layers. In traditional

convolutional networks, separate kernels are applied to each input channel, which can lead to inefficiencies and unnecessary computational complexity, particularly when working with one-dimensional vibration signals, as is common in rotating machinery diagnostics. These signals typically exhibit low channel variability, meaning the important features do not necessarily differ significantly across multiple channels. By applying a single set of kernels across all channels, the CSSUCLayer eliminates the redundancy found in conventional convolutional filters, ensuring that only the essential features are extracted, while also reducing the computational load. Moreover, when the neural network performs forward propagation, subsequent multiple channels are derived from the channel of the input and it is well known that there is inter-channel redundancy inside convolutional filters [47]. Given this situation, during CSSUCLayer computation, all channels share one set of kernels. To be specific, for an input X with a shape of $H \times W \times C$, a set of kernels convolves within the c-th ($c = 1, 2, \ldots, C$) channel, which is called channel-sharing. The channel-sharing approach not only minimizes inter-channel redundancy but also enhances the generalization capability of the model. Since all channels share the same kernels, the model learns more compact, generalized representations, which can be particularly effective when handling limited training data. This aligns well with the nature of vibration signals, where key fault characteristics tend to be subtle and spread across multiple channels.

On the other hand, the spatial-unidirectional convolution concept takes advantage of the structured layout of vibration signals. In rotating machinery, each row in the matrix corresponds to a signal generated by the passage of a rolling element, and while there is significant similarity within each row, differences arise between rows due to the mechanical impacts as the rolling elements pass by the accelerometer. This structured difference is exploited by applying distinct kernels along the height of the input, while sharing the same kernel across the width. The specific operation is that the e-th kernel K_e convolves with the input from the $[1+(e\text{-}1)s\text{-}k/2]$ to the $[1+(e\text{-}1)s\text{-}k/2]$ rows, across all columns, where $e = 1, 2, \cdots, Nk$. This is called spatial-unidirectional. N_k represents the number of kernels, which depends on the shape of input and it can be calculated by Eq. (4). Additionally, s represents the stride, indicating the step size of kernel movement, and k represents the size of kernel, specifying the dimensions of the kernel. This design is particularly advantageous for capturing row-wise variations that are indicative of faults or anomalies. By allowing the kernels to vary across different rows, the CSSUCLayer can capture the subtle changes in the signal as each rolling element interacts with the machinery. This selective convolution mechanism provides a fine-grained analysis of row-wise features, which enhances the model's ability to detect local anomalies in the vibration signal. The two operations mentioned above are collectively referred to as channel-sharing and spatial-unidirectional convolution.

$$Nk = \frac{H\text{-}k}{s} + 1 \tag{4}$$

To establish the correlation of signals across different time periods, two-dimensional kernels are applied. The computational process of CSSUCLayer is illustrated as follows:

$$Y_{i,\,j,\,c} = \sum_{m=1}^{k} \sum_{n=1}^{k} X_{is+m\text{-}1,\,js+n\text{-}1,\,c} K_{e,\,m,\,n} \tag{5}$$

where Y represents the output, (i, j, c) represents the corresponding coordinate of the output, and (m, n) represents the coordinate index within each kernel. The output at height position index i is obtained by convolving the e-th kernel with the input at height position index $is + m\text{-}1$, so i equals to e. Under this computational principle, the number of kernels is equal to the height of the output.

The kernels of CSSUCLayer are generated conditioned on the input as illustrated in Fig. 4, where * denotes the convolution operation. Firstly, a convolutional layer with kernel size of 1×1, output channel of 1, and stride of 1 is applied. Secondly, a convolutional layer with kernel size of 1×1, output channel of k^2, and stride of 1 is used to adjust the number of channels. The first two steps are referred to as a bottleneck layer, and through this bottleneck layer, an output of $H \times W \times k^2$ is obtained. Using this method to generate kernels can better extract features from the input. Then, an average pooling layer with kernel size of $k \times W$ and stride of s is used to reduce the width of the input to 1 and the height of the input is adjusted to $(H\text{-}k)/s + 1$, which is represented as H_{out}. Finally, the output is reshaped to $H_{out} \times k \times k$, where H_{out} is equal to N_k. Following these steps, the kernels are generated.

From a computational perspective, the CSSUCLayer improves efficiency by reducing the dimensionality of the input while retaining critical spatial relationships. The two-stage kernel generation process, where a bottleneck layer is first applied followed by an average pooling layer, ensures that the number of parameters remains manageable. The bottleneck layer compresses the channel dimension, focusing on the most important features, while the pooling layer reduces the spatial dimensions in a controlled manner. This process not only optimizes the

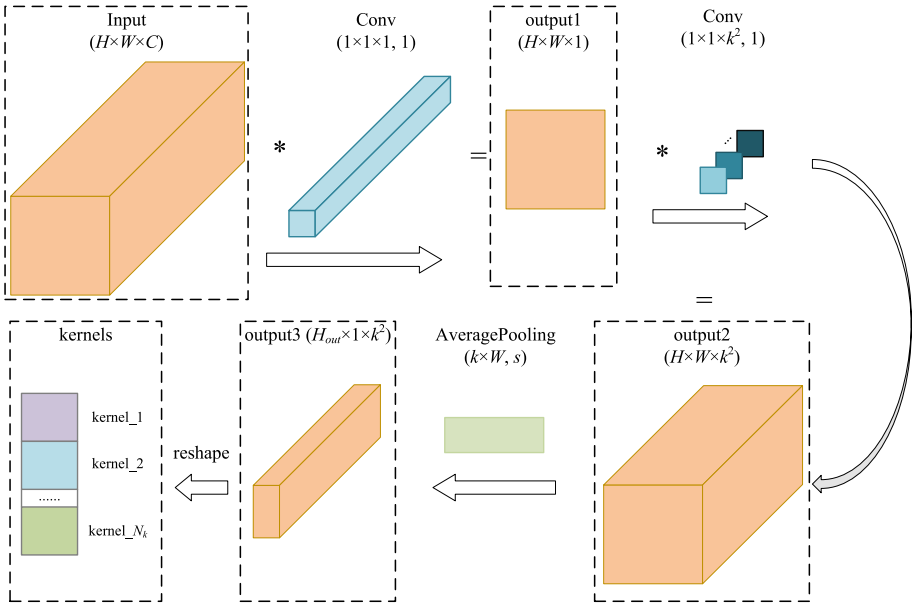


Fig. 4. Schematic diagram of the process of generating kernels.

computational load but also enhances the feature extraction capability. By reshaping the input to a more compact representation, the CSSUCLayer captures the core features necessary for accurate fault diagnosis, without being overwhelmed by the noise or irrelevant details in the input signal. The combination of bottleneck and pooling layers allows the model to maintain a balance between precision and computational efficiency, making it well-suited for real-time diagnostic applications.

The number of parameters of CSSUCLayer can be calculated as follows:

$$\mathrm{Paras_{CSSU}} = \mathrm{H}_{out}\mathrm{k}^2 \tag{6}$$

As the kernels are generated through a bottleneck layer, the number of trainable parameters actually refer to the parameters of the bottleneck layer, which can be calculated by Eq. (7).

$$\mathrm{Paras_{T\text{-}CSSU}} = \mathrm{C} + \mathrm{k}^2 \tag{7}$$

The floating point operations (FLOPs) of CSSUCLayer are composed of two part, one of which is bottleneck layer to generate kernels, and the other is convolution operation between the kernels and the input. The FLOPs of CSSUCLayer can be calculated by Eq. (8).

$$\mathrm{FLOPs} = 2\mathrm{CHW} + 2\mathrm{k}^2\mathrm{HW} + 2\mathrm{k}^2\mathrm{CH_{out}W_{out}} \tag{8}$$

where W_{out} represents the width of the output.

2.2.2. CSSUCNet

As described above, CSSUCLayer is a neural network layer. Therefore, it can replace convolutional layers in any network framework, exhibiting strong versatility.

This configuration not only facilitates the adjustment of channel numbers but also significantly reduces the model's parameters and FLOPs compared to placing these layers before the CSSUCLayer with a stride of 2.

In this paper, CSSUCNet, illustrated in Fig. 2(c), is constructed using ResNet18 [48] as the backbone framework. In this architecture, convolutional layers are substituted with CSSUCLayers, except for those in the residual blocks. Since there is only one set of kernels for CSSUCLayer, it cannot change the number of input channels. To address this limitation, four additional convolutional layers with kernel size of 1×1 and stride of 1 are added after CSSUCLayer that operates with a stride of 2. This configuration not only facilitates the adjustment of channel numbers but also significantly reduces the model's parameters and FLOPs compared to placing these layers before the CSSUCLayer.

3. Experimental procedure

3.1. Dataset description

3.1.1. Case 1: Paderborn University bearing dataset

Recently, the rolling bearing dataset of Paderborn University [49] has been widely used as the benchmark bearing fault data. The experimental subject is 6203 ball bearing with real damages caused by accelerated lifetime test and an accelerometer with 64000 Hz sampling frequency is installed on the housing to collect vibration signals. The diameter of the balls revolving around the inner ring and the diameter of the ball are 28.55 mm and 6.75 mm respectively. The contact angle is 0. The number of balls is 8. There are four different working conditions, and the basic setup with rotational speed of the inner ring $\mathrm{n} = 1500$ rpm, a load torque of $\mathrm{M} = 0.7$Nm and a radial force on the bearing of $\mathrm{F} = 1000$ N is selected to evaluate the proposed method. According to the fault location, there are three types: healthy, inner ring fault and outer ring fault, so the number of labels in the experiment is three, consistent with the dataset provider in reference [49]. The number of data points W_0 can be obtained from Eq. (1) to (3). The final bearing fault diagnosis dataset information is shown in Table 1.

Table 1
Paderborn University bearing dataset.

Type	Training set	Testing set	Shape ($N \times W_0 \times 1$)	Label
Healthy	2800	700	$8 \times 839 \times 1$	0
Inner ring fault	2800	700	$8 \times 839 \times 1$	1
Outer ring fault	2800	700	$8 \times 839 \times 1$	2

3.1.2. Case 2: PRSM dataset

To further evaluate the proposed method, an experiment is conducted using a self-collected dataset on another type of rotating machinery, referred to as PRSM. The structure of PRSM is shown in Fig. 2 (a). The diameter of the rollers revolving around the screw and the nominal diameter of the roller are 32 mm and 8 mm respectively. The contact angle is 0. The number of rollers is 10. When screw rotates around its axis, the rollers distributed between the screw and nut are driven to rotate around their own axis, while the rollers revolve around the axis of the screw, driving the nut to perform linear motion. For PRSM, the outer ring refers to the nut, which is not directly fixed to the housing but leaves a certain gap. Therefore, the accelerometer can be directly installed on the nut. The PRSM testing rig, as shown in Fig. 5, is mainly composed of servo motor, coupling, tested PRSM and hydraulic loading system. There are three levels for load: 3kN, 6kN and 9kN respectively. In this paper, there are three types of tested PRSM: healthy, lubrication failure and broken teeth of roller, and they are displayed in Fig. 6.

In this experiment, an accelerometer with a sampling frequency of 20480 Hz is installed on the nut and the screw rotates at a speed of 105 rpm. To enhance the number of samples and reduce the width of the samples to an appropriate size, a down-sampling strategy with a sampling interval (si) of 7 is employed and sa-th sample is represented as sample_sa, which is illustrated in Fig. 7. The final PRSM fault dataset information is shown in Table 2. Adjacent points in each sample are spaced 7 units apart in the raw data. In this approach, the sampling frequency has decreased to one-seventh of its original rate, effectively increasing the total number of samples sevenfold.

3.2. Noise injection

The vibration data collected in industrial environments are easily polluted by noise, which will affect the diagnosis results. In order to evaluate the anti-noise ability and feature extraction capability of the proposed method in noise conditions, Gaussian white noise and impact noise are injected into data.

The SNR is usually used to measure the intensity of Gaussian white noise interference, representing the ratio of signal power to noise power in signal [37]. The definition of SNR is shown as follows:

$$\mathrm{SNR} = 10\log\frac{\mathrm{P_{signal}}}{\mathrm{P_{noise}}} \tag{8}$$

where P_{signal} and P_{noise} are the power of the signal and the noise respectively.

In this paper, the noise injection method and purpose are similar to those in references [34] and [35]. Gaussian white noise with $\mathrm{SNR} = 10$ dB, 4 dB, 0 dB and -4 dB are respectively injected into the data.

In real industrial environments, when the displacement, load, or speed of rotating machinery experiences sudden changes, such as loose components, sudden changes in external load, structural resonance, rapid start-stop operations, etc., impact will be generated in the signal. To further evaluate anti-noise ability and feature extraction capability under impact noise conditions, this paper proposes a method for injecting impact noise into the data. The injection method for impact noise is articulated as follows:

$$\mathrm{Noise_{imp}} = \mathrm{Aexp}^{-\alpha t}\cos(\beta \mathrm{t}) + \varepsilon \tag{9}$$

where A represents the amplitude of the impact noise, exp is the

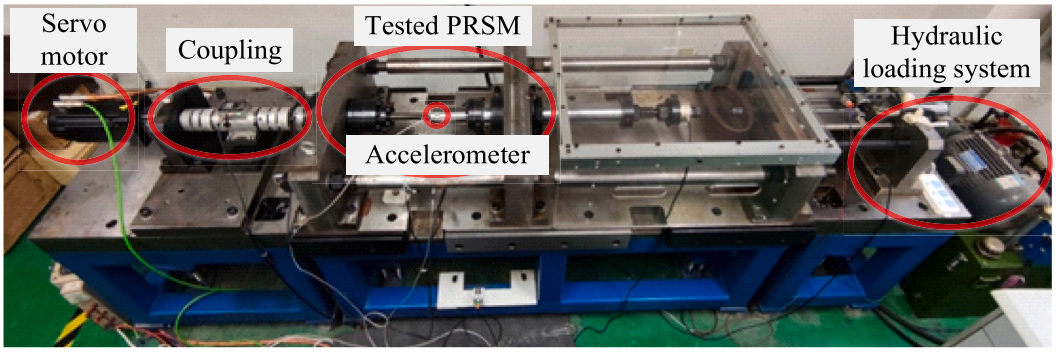


Fig. 5. The PRSM testing rig.

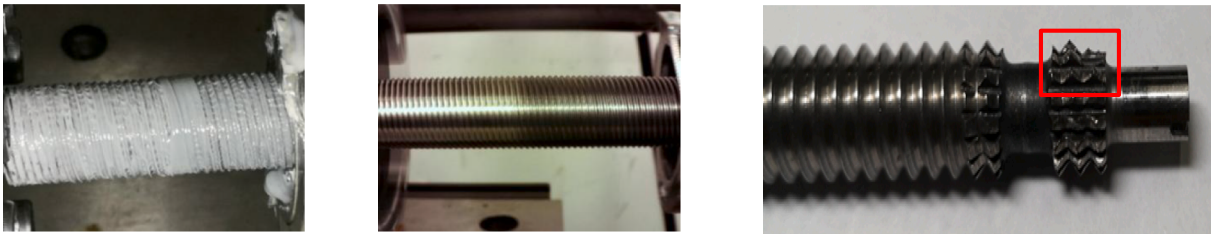


Fig. 6. Three types of tested PRSM: (a) healthy; (b) lubrication failure; (c) broken teeth of roller.

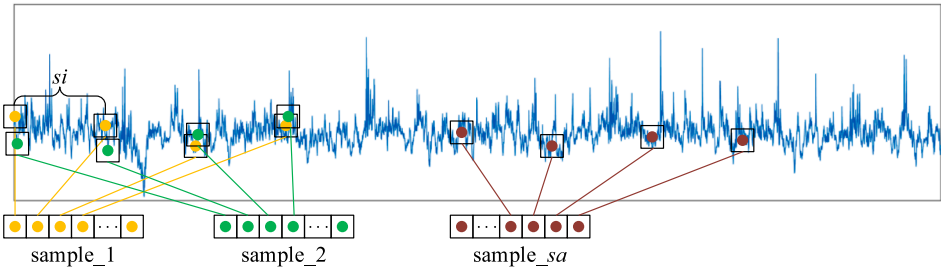


Fig. 7. Down-sampling strategy.

Table 2
PRSM dataset.

Type	Training set	Testing set	Shape ($N \times W_0 \times 1$)	Label
Healthy	252	63	$10 \times 446 \times 1$	0
Lubrication failure	252	63	$10 \times 446 \times 1$	1
Broken teeth of roller	252	63	$10 \times 446 \times 1$	2

Euler number, α represents the attenuation coefficient, which is related to the shape of the component, β represents the vibration velocity, which is related to the material of the component, t represents the time and ε represents random disturbance. Unlike Gaussian white noise, which affects all data points in a signal, impact noise influences only specific data points within a segment. When the shape and material of a component remain constant, a stronger impact or higher frequency of impacts results in a greater number of affected data points, leading to an increased proportion of data points influenced by impact noise in the signal. Currently, there is no literature on fault diagnosis that considers the interference of impact noise on model performance, leaving a lack of standard references for injecting such noise. To quantify the intensity of impact noise interference, this study introduces a metric denoted as PRO, representing the proportion of data points affected by impact noise within the signal. A larger PRO indicates a greater amplitude or higher frequency of impact noise. To more accurately simulate real industrial scenarios, random disturbance is also incorporated into the impact noise. Furthermore, the locations where impact noise occur in the signal are arbitrary. In this paper, impact noise with PRO = 5 %, 10 %, 15 % and 20 % are injected into the data in this paper. In order to demonstrate the impact noise more clearly, the raw signal and the signal containing impact noise with PRO = 20 % are shown in Fig. 8.

3.3. Experimental setup

During the experiment process, the training set is utilized to train the model and the testing set is used to test the performance of the model. The testing set is not involved in model training. In the training process, the Adam optimizer is used. In order to mitigate overfitting, learning rate adjustment strategy is employed. The maximum number of epochs is set to 100. The initial learning rate is set to 0.01, and learning rate is reduced to 0.1 times its previous value at epochs 10, 20, 25, 50, and 75. Each experiment is repeated four times and the mean and standard deviation are used as the final experimental results to evaluate the ability of each method.

In order to evaluate the performance of the proposed RES-CSSUC method, the comparative experiment with WKN [31], CCQNet [32], LiConvFormer [33], MobileNetV3 [29], EfficientNetV2 [30] ResNet18 [48], RES-ResNet18, and RES-CSSUC-R is carried out, where the descriptions of these compared methods are as follows:

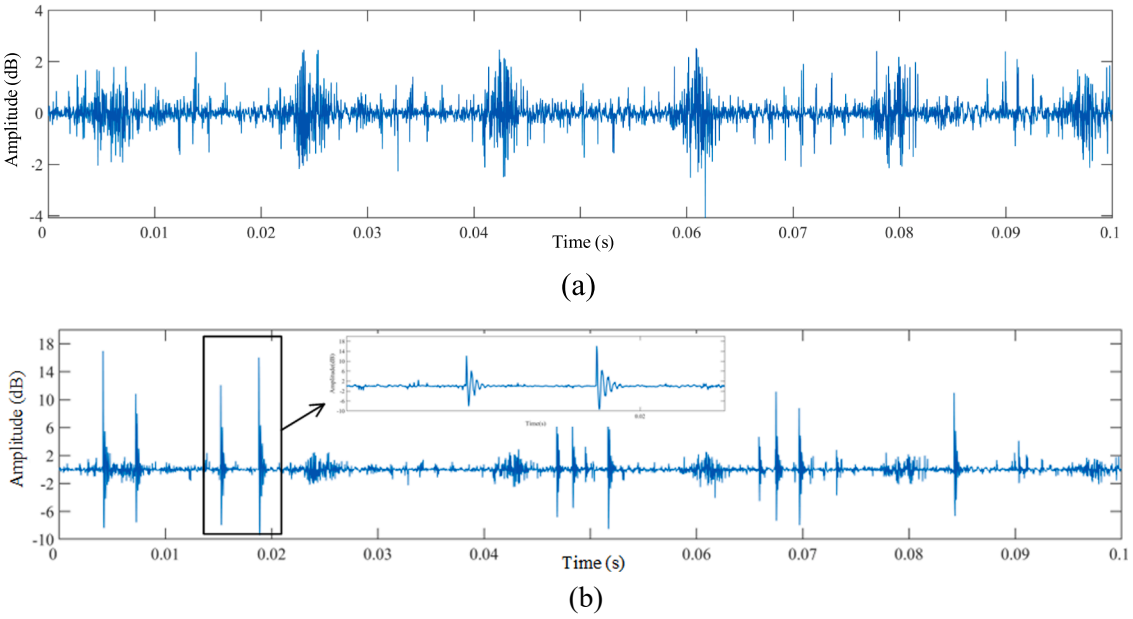


Fig. 8. Vibration signal: (a) raw vibration signal; (b) vibration signal containing impact noise with PRO = 20 %.

(1) WKN, a fault diagnosis method, the input sample of which is a one-channel one-dimensional matrix with sample length L, which corresponds to the number of data points collected by the accelerometer when inner ring rotates one revolution around its axis. This comprehensive sampling captures the complete working cycle, ensuring thorough information. Additionally, during the operation of rotating machinery, the motion is typically cyclical, which enables comparative analysis across different time windows. Moreover, using one revolution data ensures consistency between samples, enhancing model stability and reliability during training and inference. The backbone framework of WKN is ResNet18.

(2) CCQNet, a fault diagnosis method, the input sample of which is a one-channel one-dimensional matrix and the sample length L is the same as that of WKN. Its backbone framework is ResNet18.

(3) LiConvFormer, a fault diagnosis method, the input sample of which is a one-channel one-dimensional matrix and the sample length L is the same as that of WKN. LiConvFormer consists of a total 29 neural network layers.

(4) MobileNetV3, a lightweight model, the input sample of which is a one-channel two-dimensional matrix. The signal is not processed using RES data processing technique. Instead, similar to most fault diagnosis methods, the one-dimensional signal is represented as a one-channel two-dimensional matrix with a shape of $\sqrt{L} \times \sqrt{L} \times 1$, where L represents the number of data points collected by the accelerometer when inner ring rotates one revolution around its axis and the reason for selecting L data points as a sample has been explained above. MobileNetV3 consists of a total of 54 neural network layers.

(5) EfficientNetV2, a lightweight model, the input sample of which is a one-channel two-dimensional matrix and the signal is processed with the same method as that of MobileNetV3, so the shape of the input sample is $\sqrt{L} \times \sqrt{L} \times 1$. EfficientNetV2 consists of a total of 37 neural network layers.

(6) ResNet18, a lightweight model, the input sample of which is a one-channel two-dimensional matrix and the sample is processed with the same method as that of MobileNetV3. The shape of the input sample is $\sqrt{L} \times \sqrt{L} \times 1$.

(7) RES-ResNet18, in which ResNet18 is used as the neural network model. The input sample is processed with RES signal processing technique, so the shape of sample is the same as that of the proposed method.

(8) RES-CSSUC-R, in which the input sample mirrors that of the proposed method. This method reverses the principle of CSSUCLayer in the spatial extent, where different rows share the same kernel, while the kernel differs across different columns. Assuming that the signal between every two adjacent rolling elements forms a segment, this method is intended for further evaluating that there is no significant change in the characteristics within each segment of the signal, rather than assuming similarity between the characteristics of every two segments of the signal.

From the above introduction, it can be seen that the comparison models with the same type of input as the method proposed in this paper include RES-ResNet18 and RES-CSSUC-R, both of which incorporate the RES data processing technique, demonstrating the role of the proposed CSSUCNet. WKN, CCQNet, and LiConvFormer are selected as comparison models for high-performance models in the fault diagnosis domain. The original one-dimensional signals are processed of these three methods, but based on their own proposed methods, the signals remain one-dimensional after processing. MobileNetV3, EfficientNetV2, and ResNet18 are chosen as comparison models for high-performance lightweight models. For MobileNetV3, EfficientNetV2, and ResNet18, since no signal processing was applied, this paper uses a two-dimensional matrix, which is the same input format as that of CSSUCNet, as the input to the neural network. Essentially, the original input for all these methods (including the proposed method in this paper) is a one-dimensional vibration signal.

4. Results and discussions

To more clearly compare the proposed method with other high-performance fault diagnosis methods and lightweight models, as well as to highlight the contribution of each module of the proposed method, the ablation study and performance comparison experiments are conducted simultaneously. To be specific, the ablation study of ResNet18 and RES-ResNet18 verifies the contributions of the proposed RES data processing technique. The ablation study of RES-ResNet18 and RES-CSSUC validates the effectiveness of the proposed CSSUCLayer and CSSUCNet. Additionally, the ablation study between RES-CSSUC-R and RES-CSSUC validates the theoretical hypothesis presented in Section 2.1, specifically that there is no significant change in the signal characteristics between two adjacent rolling elements.

4.1. Case 1: The results and discussions of Paderborn University bearing dataset

4.1.1. Effectiveness

Table 3 presents the mean and standard deviation of accuracy for each method, along with the accuracy of the best-performing model, i.e., the highest accuracy among the four repeated trials. Fig. 9 and Fig. 10 display the confusion matrices and the t-SNE results of the best-performing model, respectively. As indicated in Table 3, RES-CSSUC, RES-ResNet18, and RES-CSSUC-R achieved the top three accuracies, with 99.99 %, 99.99 %, and 99.91 %, respectively, among which the signals are preprocessed with RES signal processing technique. The performance of these three methods significantly surpasses that of other methods not employing RES signal processing technique, demonstrating the effectiveness of RES signal processing technique and confirming that the signal characteristics between adjacent rolling elements do not significantly change. In Fig. 9, it is evident that all other methods correctly classify the healthy samples, highlighting the substantial differences between the characteristics of healthy and faulty samples. However, most methods struggle to accurately classify samples of inner ring fault and outer ring fault. likely due to their similar characteristics—such as fault characteristic frequency, vibration characteristic and fault progression. This failure suggests that most methods do not effectively extract subtle differences in features among samples from different classes. Fig. 10 further reveals a clear classification boundary between the healthy and faulty samples. However, there is partial overlap between the samples of inner ring fault and outer ring fault for most methods. While CCQNet achieved a highest accuracy of 100 %, its average accuracy is lower and its standard deviation is higher compared to RES-CSSUC and RES-ResNet18, indicating that the stability of CCQNet is not as good as that of RES-CSSUC and RES-ResNet18. Additionally, the t-SNE results for RES-ResNet18 and RES-CSSUC-R shown in Fig. 10(g) and (h) exhibit greater dispersion, while the t-SNE result for RES-CSSUC shown in Fig. 10(i) is more concentrated within each class, suggesting clearer classification boundaries among different classes.

4.1.2. Interpretability research

To explore how the CSSUCLayer differentiates among various fault characteristics, a detailed explanation of how the learned features correspond to different types of faults in rotating machinery is proposed. This can be done by analyzing the feature maps produced by the CSSUCLayer and correlating them with known fault signatures from the data. To finish this task, Gradient-weighted Class Activation Mapping (Grad-CAM) is deployed to enhance the model's interpretability, particularly for fault diagnosis tasks. Grad-CAM can provide valuable visualizations of which parts of the input data are most influential for the model's predictions, which can bridge the gap between model outputs and physical fault characteristics.

First, Fig. 11 presents the output matrix of the RES technique, where Fig. 11(a) shows data from the same type but different rows, and Fig. 11 (b) shows data from different types. It can be observed that after the FFT transformation, each row of data contains a prominent peak, which is likely caused by the rolling elements passing the accelerometer. However, the location of the peak varies across different rows, possibly due to changes in the signal characteristics after the rolling elements pass the accelerometer. For signals from different states, the location of the maximum frequency amplitude is almost the same, but the frequency components differ significantly. For example, for healthy bearings, there are frequency components greater than the frequency with the maximum amplitude, but with amplitudes smaller than the amplitude at the frequency with the maximum amplitude. However, for inner ring fault and outer ring fault, the main frequencies contain both components greater than the frequency with the maximum amplitude and those smaller than the frequency with the maximum amplitude. However, the frequency and amplitude of the two states are different.

To explain how the layer differentiates among various fault characteristics, this paper conducts a Grad-CAM visualization analysis on the data from three bearing conditions and the network weights, as shown in Fig. 12.

As shown in Fig. 12, CSSUCNet not only focuses on the index of the position where the maximum frequency amplitude occurs (highlighted in the red box in the figure), but also pays attention to the index of the position where frequency components exhibit differences (highlighted in the yellow box). The position of the maximum frequency amplitude reflects different fault types due to variations in amplitude, while the locations with differences in both frequency and amplitude are more effective in distinguishing between different fault types. This corresponds well with the frequency-amplitude plot shown in Fig. 11(b). From the heatmap, it can be observed that although the critical positions that contribute to classification decisions are generally the same across different rows of sample, the importance of these positions in determining the classification result varies. This suggests that different kernels are needed to extract distinct features between different rows, which further validates the correctness of using different kernels for different rows. This observation highlights the necessity for the model to employ diverse kernels to capture the unique characteristics of each row, thereby enhancing the model's ability to differentiate between varying fault types. This supports the design choice of using distinct kernels for different rows to improve the overall performance and interpretability of the model.

To further analyze the mechanism of fault feature extraction by different channels of CSSUCNet, Fig. 13 illustrates the Grad-CAM visualization of different channels of one sample. As shown in Fig. 13, the position indices that play a crucial role in classification decisions across different channels are largely the same and correspond to the frequency components presented in Fig. 11(b). This further supports the idea that there is feature redundancy between the channels, providing evidence for the rationale behind the CSSUC layer's design, where multiple channels share a set of kernels. This redundancy suggests that the model can efficiently capture the most relevant features across channels, reinforcing the effectiveness of this shared kernel approach. By focusing on the same critical positions of different channels, the model demonstrates its ability to extract consistent and meaningful fault features, leading to improved fault classification capability.

4.1.3. Model complexity

In this paper, as shown in Table 4, the number of parameters of each model is compared, and in order to assess the complexity of the models, the FLOPs of each model are also listed for comparison. In actual industrial scenarios, what engineers are more concerned about are the occupied memory and inference time of the model, which are also listed in Table 4. During model training, two GeForce GTX 1080 Ti GPUs are utilized. However, most industrial computers use the CPU as the computing unit. During model validation, the computing power was constrained. Model validation was conducted using a Core (TM) i5-10200H CPU to simulate a resource-limited environment. The inference time refers to the time taken for calculations using the CPU. It is also worth mentioning that the inference time refers to the total inference time for all samples, including samples loading, model loading, and

Table 3
The diagnostic accuracy of all methods on Paderborn University bearing dataset.

Method	Diagnostic accuracy (%)	Maximum accuracy (%)
WKN	99.55 ± 0.12	99.71
CCQNet	99.62 ± 0.35	100.00
LiConvFormer	99.83 ± 0.06	99.90
MobileNetV3	99.61 ± 0.08	99.71
EfficientNetV2	99.54 ± 0.13	99.67
ResNet18	99.55 ± 0.04	99.57
RES-ResNet18	99.99 ± 0.02	100.00
RES-CSSUC-R	99.91 ± 0.07	100.00
RES-CSSUC	99.99 ± 0.02	100.00

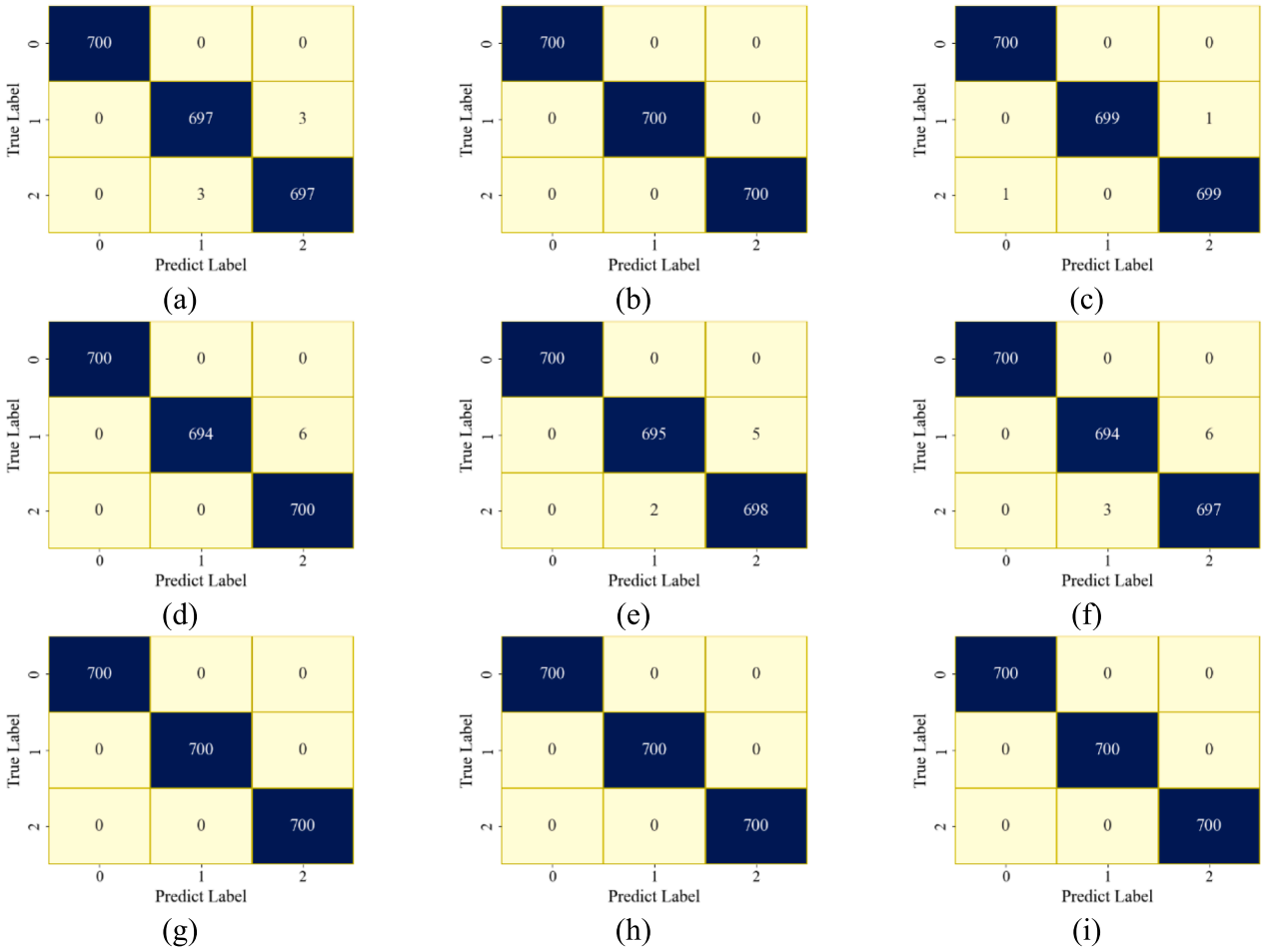


Fig. 9. The confusion matrices of all methods on Paderborn University bearing dataset: (a) WKN; (b) CCQNet; (c) LiConvFormer; (d) MobileNetV3;(e) EfficientNetV2; (f) ResNet18; (g) RES-ResNet18; (h) RES-CSSUC-R; (i) RES-CSSUC.

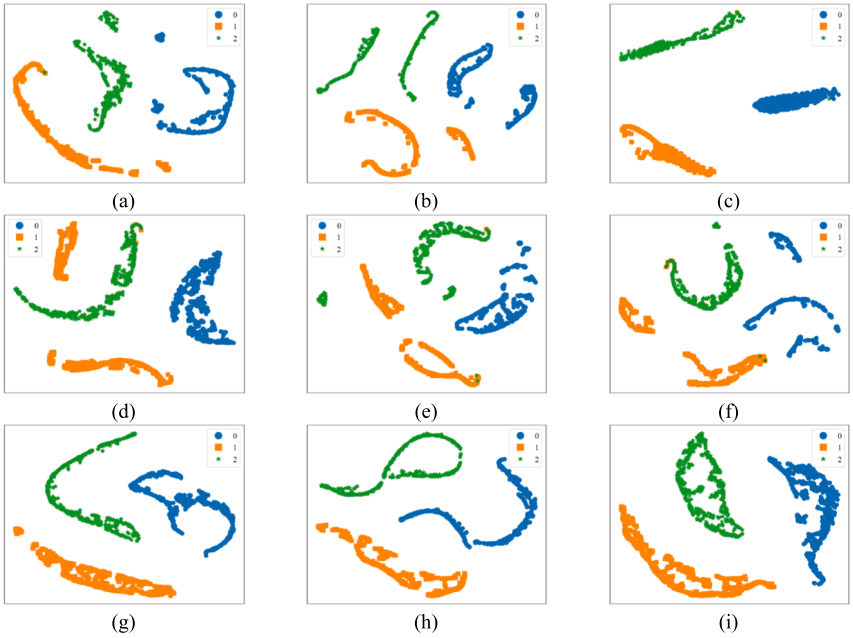


Fig. 10. The t-SNE results of all methods on Paderborn University bearing dataset: (a) WKN; (b) CCQNet; (c) LiConvFormer; (d) MobileNetV3;(e) EfficientNetV2; (f) ResNet18; (g) RES-ResNet18; (h) RES-CSSUC-R; (i) RES-CSSUC.

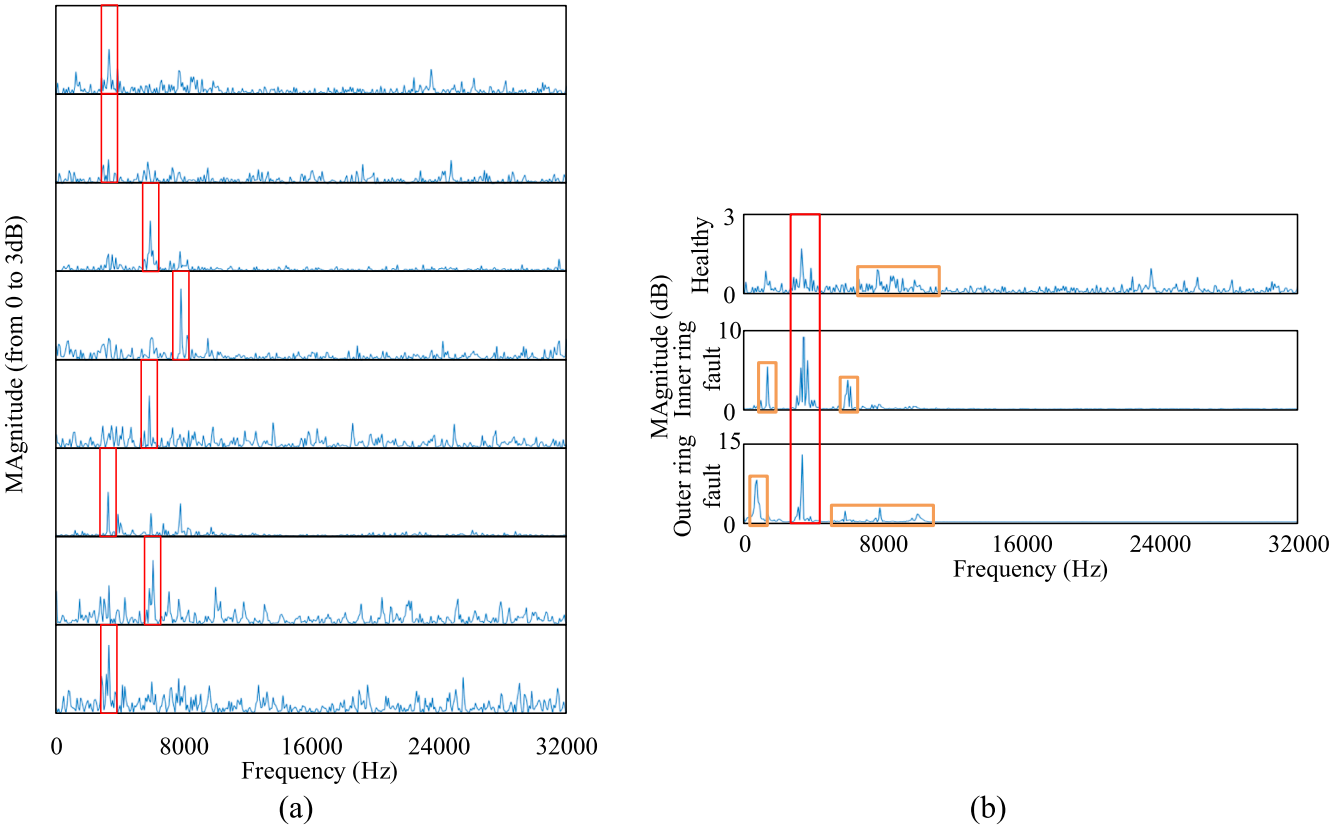


Fig. 11. The input of CSSUCNet: (a) data from the same type but different rows; (b) data from different types.

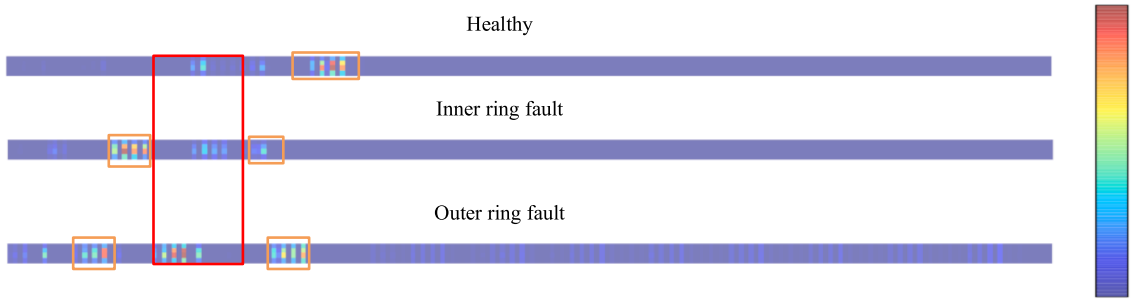


Fig. 12. Grad-CAM visualization of different types of bearings.

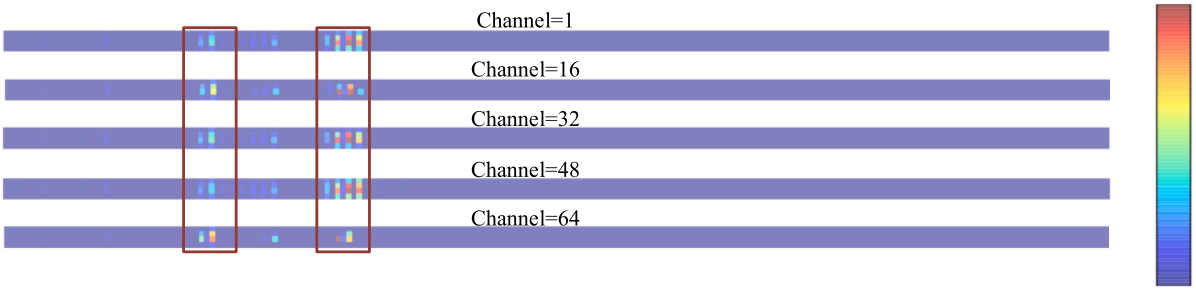


Fig. 13. Grad-CAM visualization of different channels of one sample.

inference. Each method is tested four times, and the mean and standard deviation of inference time are provided in Table 4. With identical inputs and backbone framework, RES-CSSUC has 96.78 % fewer parameters, 96.53 % fewer FLOPs, 96.49 % less memory consumption, and 19.00 % faster inference speed compared to RES-ResNet18. Despite the significant advantages in number of parameters and FLOPs, the inference time of RES-CSSUC is comparable to that of ResNet18. The possible reason is that each CSSUCLayer in RES-CSSUC requires to generate kernels, and the process is sequential, making parallel computation infeasible. For MobileNetV3, which also utilizes two-dimensional

Table 4
Model complexity of all methods.

Method	Parameters (M)	FLOPs (M)	Occupied memory (MB)	Inference time (s)
WKN	3.849	887.969	14.76	325.10 ± 0.15
CCQNet	0.726	167.933	2.94	15.18 ± 0.66
LiConvFormer	0.321	36.756	1.25	6.47 ± 0.33
MobileNetV3	1.673	5.791	6.55	7.34 ± 0.06
EfficientNetV2	20.316	201.714	78.38	175.48 ± 0.16
ResNet18	11.172	94.821	42.70	19.53 ± 0.03
RES-ResNet18	11.172	427.392	42.70	23.85 ± 0.16
RES-CSSUC-R	0.359	14.842	1.50	19.15 ± 0.10
RES-CSSUC	0.360	14.843	1.50	19.32 ± 0.04

convolutional layers, it has 78.48 % more parameters compared to RES-CSSUC. This is due to MobileNetV3 having a deeper framework with 54 layers. However, despite being deeper, MobileNetV3 achieves fewer FLOPs, less occupied memory, and faster inference speed. This is because MobileNetV3 includes 41 layers with 1 × 1 convolutional kernels, which significantly reduces the computational complexity of the model. LiConvFormer has 29 layers and introduces many multiplication and addition operations, so it does not have an advantage in terms of FLOPs compared to RES-CSSUC. However, its number of parameters, occupied memory, and inference speed are superior to RES-CSSUC. The possible reason is that LiConvFormer includes 19 layers with 1 × 1 convolutional kernels, which can greatly reduce the number of parameters and FLOPs. Moreover, both the input and the convolutional layers are one-dimensional, providing it with greater advantages in terms of model complexity and inference speed compared to RES-CSSUC. Apart from LiConvFormer, CCQNet also shares this characteristic, resulting in faster inference speed compared to RES-CSSUC.

In summary, RES-CSSUC facilitates deployment on edge device, reducing storage and computational costs. Beyond these advantages, RES-CSSUC is also beneficial for the field of federated learning proposed by Google Inc. in 2017 [50], which reduces the time of parameters uploading, aggregating, and returning.

4.1.4. Performance under Gaussian white noise and impact noise conditions

In order to simulate the external interference present in real industrial scenarios and evaluate feature extraction capability of all methods in noise conditions, Gaussian white noise and impact noise are respectively injected into data. It can be found from Table 5, the diagnostic accuracy decreases as the level of Gaussian white noise or impact noise interference increases, while the standard deviation generally rises. This indicates that both the overall performance and stability of the models degrade as the noise level increases. Notably, RES-CSSUC and RES-ResNet18 maintain high diagnostic accuracy under varying Gaussian white noise conditions (10 dB, 4 dB, 0 dB and −4 dB). Even when SNR = -4 dB, RES-ResNet18 and RES-CSSUC can achieve diagnostic accuracy of 99.56 %, while when SNR = 4 dB and 0 dB, RES-CSSUC slightly outperforms RES-ResNet18 in terms of diagnostic accuracy.

The performance of other methods significantly degrades under Gaussian white noise conditions. Methods without the RES signal processing technique show significant performance degradation under Gaussian white noise, confirming the robustness of RES-based methods. Specifically, RES-CSSUC, RES-CSSUC-R, and RES-ResNet18, by leveraging the RES signal processing technique, demonstrate superior noise resistance compared to methods without it. This provides clear evidence that the RES signal processing technique is effective in mitigating Gaussian white noise interference. Furthermore, RES-ResNet18 transforming one-dimensional signals into two-dimensional matrices based on the characteristics of rotating mechanical structures offers better interpretability and noise resistance compared to simple transformations used in other methods, such as the standard ResNet18. One of the key reasons the RES signal processing technique performs well in noise conditions is its alignment with the natural mechanical properties of rotating machinery. Unlike traditional time-domain methods, which are sensitive to both signal and noise, RES signal processing technique restructures the vibration signal to emphasize the repetitive nature of the mechanical interactions. By dividing the signal into sections corresponding to the angular displacement of the rolling elements ($360^{\circ}/N$), the technique filters out noise that doesn't match this periodic structure. Applying FFT to the matrix rows further reveals hidden periodic components, distinguishing them from noise in the time domain. RES is robust against noise because it isolates the consistent, periodic mechanical signal, allowing it to separate meaningful fault signatures from random, non-repetitive noise. Upon closer examination, RES-CSSUC performs better than RES-CSSUC-R under noise conditions. Assuming that the signal between every two adjacent rolling elements forms a segment, the signal characteristics change when a rolling element passes through the accelerometer's installation position, whereas within each segment, the characteristics remain relatively stable. For signals with longer intervals, such as those in the first and last rows of the two-dimensional matrix, significant variations in signal characteristics occur due to the increased time gap between impacts. In these cases, the spatial domain processing used in RES-CSSUC-R, which applies the same kernel to each row, fails to accurately capture the feature differences across rows. This results in poorer performance in comparison to RES-CSSUC, which adapts its kernel usage for different rows.

In terms of performance under impact noise conditions, most methods, except for CCQNet, demonstrate relatively high diagnostic accuracy. Notably, RES-CSSUC, RES-ResNet18, and LiConvFormer perform well, achieving diagnostic accuracies of 99.86 %, 99.89 %, and 99.68 %, respectively, even when the proportion of data points occupied by impact noise in the signal is 20 %. However, LiConvFormer's poor performance under Gaussian white noise suggests that its robustness to varying types of noise is less consistent. In the presence of impact noise, CCQNet has completely lost its ability to extract features, with a diagnostic accuracy of approximately 33 %, making it inadequate for performing fault diagnosis tasks in real industrial scenarios.

Additionally, it's worth noting that most methods achieve higher diagnostic accuracy under impact noise conditions compared to Gaussian white noise. This can be attributed to the intermittent nature of

Table 5
The experimental results of all methods on Paderborn University bearing dataset under eight kinds of noise conditions.

Method	Gaussian white noise (SNR)				Impact noise (PRO)			
	10 dB	4 dB	0 dB	−4 dB	5 %	10 %	15 %	20 %
WKN	99.45 ± 0.20	98.05 ± 0.58	96.16 ± 0.26	92.85 ± 0.38	99.36 ± 0.21	99.28 ± 0.26	99.23 ± 0.21	99.08 ± 0.22
CCQNet	99.01 ± 0.76	98.74 ± 0.62	97.17 ± 1.53	94.01 ± 1.84	33.51 ± 0.04	33.46 ± 0.07	33.46 ± 0.03	33.45 ± 0.10
LiConvFormer	99.67 ± 0.19	99.48 ± 0.13	98.56 ± 0.29	95.88 ± 0.31	99.77 ± 0.05	99.75 ± 0.05	99.75 ± 0.11	99.68 ± 0.08
MobileNetV3	99.29 ± 0.16	98.39 ± 0.26	96.69 ± 0.22	93.32 ± 0.54	99.57 ± 0.10	99.20 ± 0.16	99.16 ± 0.04	98.86 ± 0.28
EfficientNetV2	98.75 ± 0.32	97.97 ± 0.48	96.77 ± 0.15	92.31 ± 0.91	98.85 ± 0.55	97.79 ± 0.72	97.48 ± 0.69	97.35 ± 1.16
ResNet18	98.82 ± 0.28	97.44 ± 0.11	95.75 ± 0.20	91.75 ± 0.82	98.92 ± 0.33	98.68 ± 0.22	98.31 ± 0.31	98.11 ± 0.33
RES-ResNet18	99.98 ± 0.03	99.89 ± 0.04	99.83 ± 0.09	99.56 ± 0.18	99.95 ± 0.06	99.95 ± 0.00	99.90 ± 0.07	99.89 ± 0.02
RES-CSSUC-R	99.56 ± 0.79	99.27 ± 0.77	98.36 ± 1.84	97.42 ± 1.54	99.75 ± 0.19	99.17 ± 0.53	99.12 ± 0.42	99.07 ± 0.53
RES-CSSUC	99.98 ± 0.05	99.92 ± 0.07	99.85 ± 0.02	99.56 ± 0.18	99.96 ± 0.05	99.95 ± 0.00	99.93 ± 0.03	99.86 ± 0.09

impact noise, which only affects specific segments of the signal, while Gaussian white noise continuously interferes with the entire signal. Nevertheless, since impact noise is prevalent in real industrial scenarios, it must still be carefully considered in the model's design and evaluation.

The combined effectiveness of the RES signal processing technique and CSSUCNet in resisting noise can be attributed to their complementary approaches in signal representation and feature extraction. RES transforms the raw vibration signals into a structured two-dimensional matrix that emphasizes the periodic interactions between the rolling elements and the outer ring. This structured representation inherently filters out non-periodic noise, allowing the model to focus on the consistent mechanical signals generated during operation. Furthermore, the CSSUCNet architecture employs channel-sharing and spatial-unidirectional convolution, which enhances the model's ability to capture meaningful features across the periodic segments while reducing inter-channel redundancy. By sharing kernels among channels, CSSUCNet minimizes the influence of noise that may affect individual channels differently. Additionally, the spatial-unidirectional convolution selectively utilizes different kernels for different rows of the matrix, which allows for nuanced feature extraction that adapts to variations in signal characteristics, including those introduced by noise. This combination of RES structured signal processing and CSSUCNet advanced convolutional architecture enables the model to robustly extract relevant features, making it more resilient to various forms of noise present in real-world scenarios.

Overall, RES-CSSUC demonstrates strong noise resistance and reliable feature extraction capabilities, outperforming other methods, which demonstrates the superiority of combining RES signal processing technique with CSSUCNet.

4.2. Case 2: The results and discussions of PRSM dataset

4.2.1. Effectiveness

Table 6 indicates that the performance of each method on the PRSM dataset is generally worse than that on the Paderborn University bearing dataset. One possible explanation is that the model may struggle to extract precise feature information due to fewer samples. Additionally, during PRSM motion process, the nut moves in a straight line as the screw rotates, rather than continuously rotating in one direction like bearings or gears. This results in a forward and reverse movement of the screw, with the nut and accelerometer reciprocating linearly. Consequently, the rotational speed of the screw will constantly change, leading to significant fluctuations in signal characteristics, which poses challenges to the feature extraction capabilities of models. Notably, only RES-CSSUC, RES-CSSUC-R, and RES-ResNet18 achieve extremely high diagnostic accuracy of 99.87 %, 99.47 %, and 99.74 %, respectively. Fig. 15 (g), (h), and (i) demonstrate that the samples of each class are relatively concentrated, with distinct classification boundaries between different classes. This further underscores the strong classification capability of RES-CSSUC, RES-CSSUC-R, and RES-ResNet18 across various states on PRSM dataset. These methods utilize RES signal processing technique, exhibiting strong robustness across different rotating machinery datasets. In contrast, the performance of other methods on PRSM dataset is notably poor, particularly for CCQNet. Even after training for 100 epochs, its diagnostic accuracy on the testing set remains at only 33.33 %. This is drastically different from its performance on Paderborn University bearing dataset, indicating poor robustness. Fig. 14 (b) illustrates that all samples are predicted as lubrication failure, indicating that CCQNet has completely failed on PRSM dataset. Unlike achieving diagnostic accuracy of over 99 % on Paderborn University bearing dataset. For WKN, LiConvFormer, MobileNetV3, EfficientNetV2 and ResNet18, there is a significant decrease in diagnostic accuracy on PRSM dataset, dropping by approximately 16 % to 28 %. Although these models can correctly identify the vast majority, or even all, of the healthy samples, they struggle with identifying samples belonging to the lubrication failure and broken teeth of roller. Compared to the experimental results on Paderborn University bearing dataset, the performance difference between RES-ResNet18 and ResNet18 on PRSM dataset is more pronounced. This further illustrates that the RES signal processing technique offers better interpretability and greater advantages.

Table 6
The diagnostic accuracy and training epochs of all methods on PRSM dataset.

Method	Diagnostic accuracy (%)	Maximum accuracy (%)
WKN	91.67 ± 0.66	92.59
CCQNet	33.33 ± 0.00	33.33
LiConvFormer	79.10 ± 0.91	79.89
MobileNetV3	79.37 ± 1.14	80.95
EfficientNetV2	72.62 ± 0.51	73.02
ResNet18	83.87 ± 0.68	84.66
RES-ResNet18	99.74 ± 0.53	100.00
RES-CSSUC-R	99.47 ± 1.06	100.00
RES-CSSUC	99.87 ± 0.27	100.00

Unlike the nearly 100 % accuracy achieved by each method on Paderborn University bearing dataset, significant performance disparities are observed across methods on the PRSM dataset. To more clearly compare the performance differences of each model on the training and testing set, Fig. 16 illustrates the diagnostic accuracy of all methods on the training and testing set on PRSM dataset. With the exception of CCQNet, EfficientNetV2 and ResNet18, all other methods achieve 100 % diagnostic accuracy on the training set. In contrast, CCQNet only attains a diagnostic accuracy of 33.33 % on the training set. The possible reason is that CCQNet has not converged within 100 epochs, but more epochs mean higher computational costs. Alternatively, CCQNet is extremely mismatched with the signal features of PRSM dataset, so even with more epochs, it still fails to converge. For WKN, LiConvFormer, MobileNetV3, EfficientNetV2 and ResNet18, the accuracy on the testing set is lower than that on the training set, indicating varying degrees of overfitting. Notably, these models do not exhibit significant overfitting on Paderborn University bearing dataset, likely due to the larger number of samples that better match the complexity of these models compared to the PRSM dataset. Yet, with the same model complexity, the training set of PRSM dataset contains only 9 % of the samples available on Paderborn University bearing dataset. The mismatch between insufficient training samples and high model complexity leads to the occurrence of overfitting. Although LiConvFormer and MobileNetV3 have fewer parameters and FLOPs, their deeper neural network architectures also contribute to overfitting on the PRSM dataset. While RES-ResNet18 and ResNet18 share the same backbone framework, RES-ResNet18 employs RES signal processing technique. The signal contains richer feature information and the model can extract suitable signal features, which ensures better compatibility between the input features and the model. To a certain extent, this alignment can significantly reduce the occurrence of overfitting. As depicted in Fig. 16, our proposed method, RES-CSSUS, which integrates RES signal processing technique with CSSUCNet, effectively avoids the occurrence of overfitting and demonstrates excellent performance on both PRSM dataset and Paderborn University bearing dataset. This indicates that RES-CSSUC is well-suited for both large-sample and small-sample fault diagnosis tasks, showcasing strong robustness.

4.2.2. Performance under Gaussian white noise and impact noise conditions

From Table 7, it can be seen that CCQNet consistently achieves a classification accuracy of 33.33 %, which matches the random chance for a 3-class classification task, indicating a complete failure to learn any classification knowledge on PRSM dataset. In contrast, other methods demonstrate varying performance on PRSM dataset, which appears to be more sensitive to interference from Gaussian white noise and impact noise. This susceptibility may be attributed to the limited number of data points per sample, resulting in fewer features and making it more

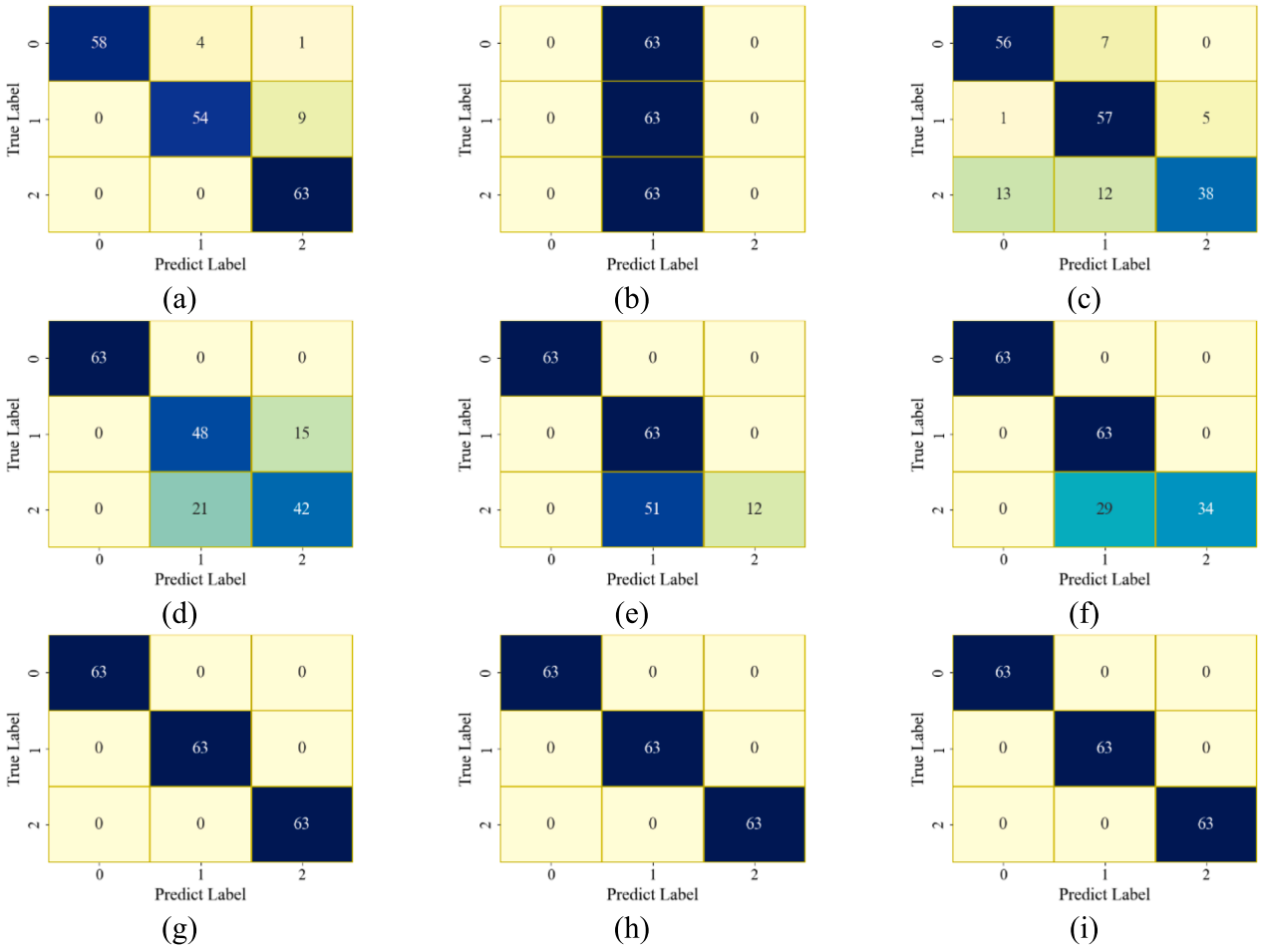


Fig. 14. The confusion matrices of all methods on PRSM dataset: (a) WKN; (b) CCQNet; (c) LiConvFormer; (d) MobileNetV3;(e) EfficientNetV2; (f) ResNet18; (g) RES-ResNet18; (h) RES-CSSUC-R; (i) RES-CSSUC.

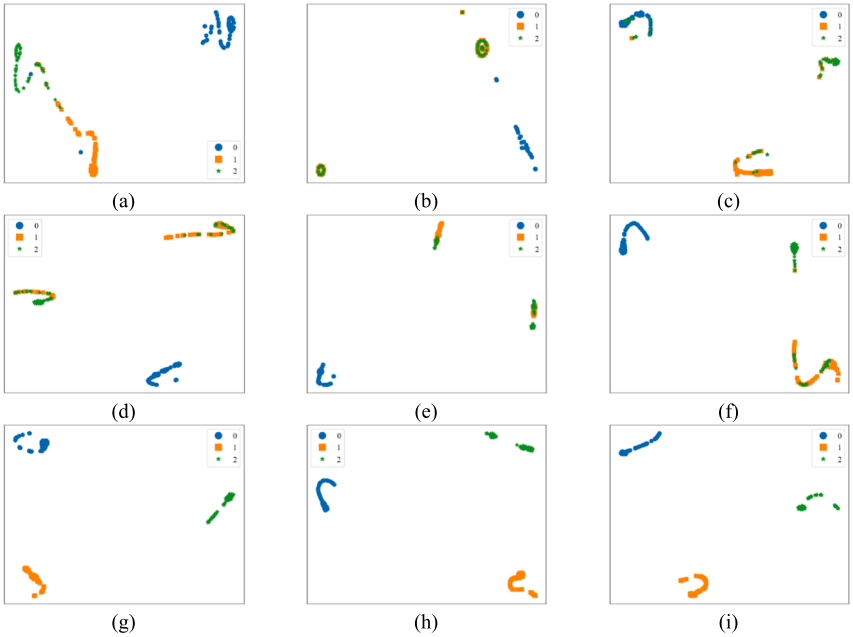


Fig. 15. The t-SNE results of all methods on PRSM dataset: (a) WKN; (b) CCQNet; (c) LiConvFormer; (d) MobileNetV3;(e) EfficientNetV2; (f) ResNet18; (g) RES-ResNet18; (h) RES-CSSUC-R; (i) RES-CSSUC.

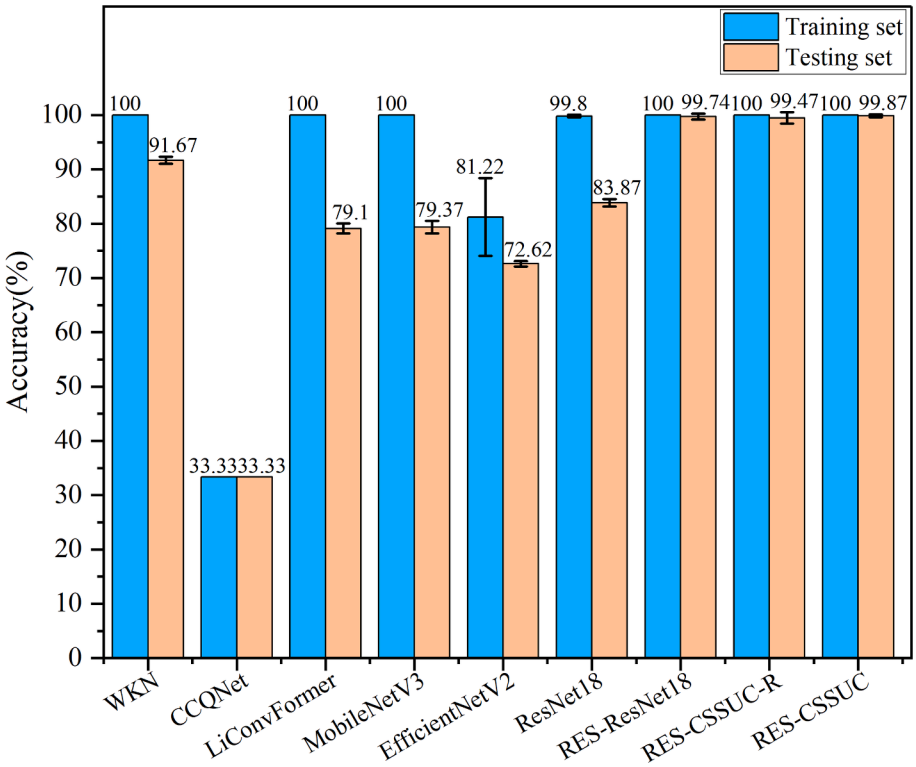


Fig. 16. The accuracy in training set and testing set of all methods on PRSM dataset.

Table 7
The experimental results of all methods on PRSM dataset under eight kinds of noise conditions.

Method	Gaussian white noise (SNR)				Impact noise (PRO)			
	10 dB	4 dB	0 dB	−4 dB	5 %	10 %	15 %	20 %
WKN	80.42 ± 0.53	80.15 ± 1.00	71.43 ± 1.78	68.43 ± 2.00	91.14 ± 1.90	89.82 ± 1.75	84.66 ± 3.12	79.10 ± 4.02
CCQNet	33.33 ± 0.00	33.33 ± 0.00	33.33 ± 0.00	33.33 ± 0.00	33.33 ± 0.00	33.33 ± 0.00	33.33 ± 0.00	33.33 ± 0.00
LiConvFormer	73.81 ± 1.75	70.51 ± 3.47	62.70 ± 5.01	48.41 ± 3.82	68.52 ± 2.88	66.80 ± 3.99	65.34 ± 3.04	62.70 ± 3.39
MobileNetV3	75.93 ± 1.76	73.81 ± 1.81	71.17 ± 2.35	70.24 ± 2.05	76.90 ± 1.33	71.69 ± 1.53	69.58 ± 1.53	69.18 ± 1.90
EfficientNetV2	66.80 ± 0.27	66.54 ± 0.27	66.54 ± 0.27	66.41 ± 0.53	67.46 ± 1.59	67.33 ± 1.00	65.74 ± 1.52	63.76 ± 4.46
ResNet18	82.14 ± 2.14	75.00 ± 2.88	69.98 ± 2.18	69.32 ± 3.08	79.37 ± 1.14	78.44 ± 1.75	78.05 ± 1.96	76.06 ± 1.39
Res-ResNet18	96.30 ± 1.29	91.54 ± 1.29	90.87 ± 2.60	88.36 ± 2.25	98.55 ± 1.58	96.03 ± 2.22	93.52 ± 2.53	91.67 ± 2.57
RES-CSSUC-R	96.43 ± 2.46	89.02 ± 2.78	82.67 ± 1.80	67.59 ± 2.17	86.64 ± 1.75	80.82 ± 1.75	80.29 ± 1.75	70.64 ± 2.54
RES-CSSUC	97.36 ± 1.49	94.84 ± 1.90	93.65 ± 0.97	93.65 ± 1.56	97.09 ± 1.40	96.16 ± 1.90	94.18 ± 0.97	93.39 ± 1.59

vulnerable to noise. Nevertheless, RES-CSSUC maintains excellent diagnostic accuracy, reaching 93.65 % even with an SNR = −4 dB, outperforming other methods by approximately 5 % to 45 % except for CCQNet. Similarly, under conditions where 20 % of the data points are impacted by impact noise, RES-CSSUC achieves a diagnostic accuracy of 93.39 %, the highest among all methods, surpassing others by 2 % to 30 % except for CCQNet. This highlights the robustness of RES-CSSUC against various noise conditions on PRSM dataset.

The strong noise resistance of RES-CSSUC can be attributed to its ability to structure the data effectively, allowing it to focus on meaningful signal patterns while filtering out random noise. This design enables the model to distinguish between the core mechanical signal and various noise types, enhancing its robustness. Additionally, the proposed method of analyzing the periodic characteristics of rolling elements helps it maintain performance despite external disturbances.

Similar to the situation on Paderborn University bearing dataset, except for CCQNet, most methods under impact noise conditions outperform the performance under Gaussian white noise conditions. Furthermore, as the SNR decreases for Gaussian white noise or as the proportion of impact noise in the signal increases, the diagnostic accuracy of all methods gradually decreases while the standard deviation shows an overall increasing trend. This is attributed to the gradual weakening of signal characteristics with the amplification of noise. As shown in Fig. 2(b) and Fig. 3, assuming the signal between the rolling elements RE_r and RE_r + 1 is represented as data_r and the signal between the adjacent rolling elements RE_r + 1 and RE_r + 2 is represented as data_r + 1. It is easy to notice that unlike the outstanding performance of RES-CSSUC, RES-CSSUC-R performs poorly on PRSM dataset, demonstrating that RES-CSSUC-R fails to extract features for segments with long-term interval using the same kernel for each row of matrix in the spatial domain. This further confirms that there is no significant change in the characteristics within each segment of the signal, such as the internal features of data_r, while significant change occurs between two segments of the signal, such as the features of data_r and data_r + 1.

5. Conclusions

To facilitate the integration of fault diagnosis system with edge device for EMA and the design of intelligent EMA, aiming to achieve condition monitoring of key transmission components, such as bearings and the PRSM, in a cost-effective and responsive manner, this paper proposes an interpretable fault diagnosis method using a signal

processing technique and a lightweight model based on mechanical structural characteristics. Firstly, leveraging the structural features of rotating machinery, the RES signal processing technique is employed to process signals and generate input samples for the subsequent lightweight model. Based on the characteristics of these input samples, CSSUCLayer is proposed, which incorporates channel-sharing and spatial-unidirectional convolution. Finally, a lightweight model, CSSUCNet, is constructed based on the backbone framework of ResNet18. The performance of the proposed method has been evaluated on Paderborn University bearing dataset and PRSM dataset. In order to assess the anti-noise ability and feature extraction effectiveness of our proposed method in noise conditions, Gaussian white noise and impact noise are injected into the signals respectively to simulate real industrial scenarios. Notably, the method of injecting impact noise into the signals is proposed for the first time. Through comparation with existing intelligent fault diagnosis methods and lightweight models, RES-CSSUC demonstrates high diagnostic accuracy while maintaining a low number of parameters, few FLOPs, and little memory usage. This results in reduced computational and storage costs, making it highly suitable for integration with mechanical equipment and advancing the development of intelligent machinery. Furthermore, RES-CSSUC can accurately extract features from the input in various noise conditions, showcasing strong anti-noise performance and reliable feature extraction ability. It exhibits excellent performance and strong robustness whether on Paderborn University bearing dataset with ample samples or on PRSM dataset with few samples. However, our research does have some limitations. One of them is the inference speed of our proposed method, which is not the fastest, primarily due to the way kernels are generated. In future work, we will focus on improving the kernel generation process to enhance the performance of our fault diagnosis method. Additionally, there is a concern regarding the model potentially becoming overspecialized to the signal characteristics of the selected datasets. While the current model has been trained with datasets representing typical machinery with well-defined structural behaviors, the importance of ensuring adaptability across different machinery types is recognized. To address this, the dataset will be expanded in the future, incorporating a wider range of operational conditions and machinery with varying structural behaviors, which will allow the model's generalization capabilities to be further evaluated and improved.

CRediT authorship contribution statement

Maodong Niu: Conceptualization, Methodology, Software, Writing – original draft. **Shangjun Ma:** Writing – review & editing, Validation, Supervision, Funding acquisition. **Haifeng Zhu:** Resources, Formal analysis. **Ke Xu:** Writing – review & editing, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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